Source:

Perseverance, NASA: [Perseverance Rover Components - NASA Science]

Product Summary:

The Perseverance rover is a NASA Mars rover that landed in Jezero Crater in 2021. Its main purpose was to study the Martian surface and search for signs of ancient microbial life. It collected rock and soil samples. The vehicle is a six-wheeled rover that uses cameras, sensors and an autonomous navigation system called AutoNav to detect obstacles, create maps of the terrarium and plan safe routes. Additionally it has a robotic arm and scientific instruments on board allowing it to examine and collect samples.

Product Critique:

While the AutoNav can handle most of the driving, humans still choose the objectives and destinations. It also has difficulty navigating the Martian terrain. Additionally, it is a science rover and not a cargo transporter. While it can carry samples and equipment it wasn't designed to move large structures. Our external cargo transport project could implement a navigation system similar to AutoNav: integrating a system that continues to update a map of the lunar surface as it moves, but the overall mission of both vehicles would be completely different.

Images/Sketches:





